From Scalar Scores to Defect-Field Geometry

Formal Definitions for Multi-Scale Structural Selection

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Abstract

Structural-selection scores are useful only after choosing a resolution. Smooth systems may admit compact scalar summaries, while graph-local, frustrated, or multi-scale systems can hide decisive information in tails, clusters, and local defect fields. This paper formalizes that distinction for MAAT v1.4.1.

We define five objects needed for a resolution-aware formulation: local defect fields $d_a(x)$, scale projections Π_ℓ, defect-field dynamics $\partial_t d_a$, observation projections Π_{obs}, and empirical universality criteria based on no-retuning transfer. Scalar structural scores are then interpreted as macro-projections of a richer defect-field geometry, not as universally sufficient descriptors.

The conservative conclusion is that universality should not be claimed at the level of a single score. At most, it may appear as partial transfer of structural modes under declared projections, shuffled nulls, and domain-shift tests.

1 Purpose and Status

The purpose of this paper is to cleanly define the operator language that was implicit in Papers 50–55. It is not a new empirical success claim and not a new physical field theory. It is a formal consolidation of the following observation:

> *Different domains require different structural resolutions. A scalar macro-score is only one projection of a richer defect-field object.*

The five definitions developed here correspond to five open points:

1. local defect-field geometry $d_a(x)$;
2. multi-scale projection and coarse-graining Π_ℓ;
3. effective dynamics of defect fields;
4. observation/projection operators Π_{obs};
5. empirical criteria for universality without retuning.

2 Structured Domains

Let a structured domain be a tuple

$$\Omega = (\mathcal{X}, \mathcal{N}, \mu, \mathcal{C}, \Phi), \tag{1}$$

where:

- $\mathcal{X}$ is the underlying set of sites, points, variables, states, or regions;
- $\mathcal{N}(x) \subseteq \mathcal{X}$ is a local neighbourhood around x;
- μ is a measure or counting weight on $\mathcal{X}$;
- $\mathcal{C}$ is a set of constraints, equations, admissibility rules, or consistency relations;
- Φ denotes the configuration data on $\mathcal{X}$.

Examples:

Domain	$\mathcal{X}$	$\mathcal{C}$
Field theory	spacetime/lattice points	equations of motion, stability
SAT	variables/clauses	clause satisfaction, propagation consistency
Cosmology	redshift/time samples	background/growth consistency
Quantum pointer tests	basis states/channels	decoherence and robustness
ML hardness	samples/feature graph	local label and manifold consistency

3 Local Defect Fields

Let $\mathcal{A}$ be the set of primary structural sectors. In the v1.2.1/v1.4.1 convention, the primitive sectors are

$$\mathcal{A} = \{H, B, S, V\}, \tag{2}$$

while robustness is treated as an emergent closure rather than an independent primitive field.

3.1 Definition

A local defect field is a non-negative local incompatibility density

$$\boxed{d_a(x) = \mathcal{D}_a\big(\Phi|_{\mathcal{N}(x)},\, \mathcal{C}|_{\mathcal{N}(x)},\, \partial\mathcal{N}(x),\, \mu|_{\mathcal{N}(x)}\big) \geq 0, \qquad a \in \mathcal{A}.} \tag{3}$$

The operator $\mathcal{D}_a$ is domain-specific, but its role is fixed: it maps local configuration data and local constraints to a comparable defect density.

3.2 Interpretation

Sector	Local defect meaning
H	local inconsistency with equations, constraints, or target relations
B	local imbalance between competing forces, signs, degrees, classes, or conservation channels
S	local activity outside the controlled useful window; too little can be stagnation, too much can be chaos
V	local disconnection, weak coupling, graph fragmentation, or loss of coherent neighbourhood linkage

The corresponding support field is

$$\Gamma_a(x) = \frac{1}{1 + d_a(x)}. \tag{4}$$

Thus high support means low local defect.

3.3 Illustrative Operationalisations

The following examples are not proposed as unique definitions. They show how the abstract operator $\mathcal{D}_a$ becomes concrete once a domain and neighbourhood rule have been declared.

For a SAT instance with clause-variable graph $G = (U \cup C, E)$ and a partial assignment A, a local consistency defect around variable $u \in U$ may be written as

$$d_H^{\mathrm{SAT}}(u; A) = \frac{1}{|\mathcal{N}(u)|} \sum_{c \in \mathcal{N}(u)} \mathbf{1}\{c \text{ is violated or unit forced under } A\}. \tag{5}$$

This is a local clause-stress density, not a global satisfiability statement. A simple connectivity defect can be defined through local graph expansion:

$$d_V^{\mathrm{SAT}}(u) = \frac{1}{\epsilon + |\mathcal{N}_k(u)|/\mathrm{median}_{v \in U} |\mathcal{N}_k(v)|}. \tag{6}$$

Variables in poorly expanding or locally isolated regions then carry larger connectivity defects.

For a labelled machine-learning dataset with samples x_i, labels y_i, and k-nearest-neighbour set $\mathcal{N}_k(i)$, a direct local consistency defect is

$$d_H^{\mathrm{ML}}(i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} \mathbf{1}\{y_j \neq y_i\}. \tag{7}$$

This is the local label-disagreement statistic used as a strong external baseline in Paper 54. A local connectivity defect may instead use the kNN radius $r_k(i)$:

$$d_V^{\mathrm{ML}}(i) = \frac{r_k(i)}{\mathrm{median}_j\, r_k(j) + \epsilon}. \tag{8}$$

Large values identify samples lying in sparse or weakly connected parts of the feature manifold.

4 From Fields to Summaries

The scalar score is recovered only after choosing summary operators. For each sector:

$$\langle d_a \rangle_\mu = \frac{\int_{\mathcal{X}} d_a(x)\, \mathrm{d}\mu(x)}{\int_{\mathcal{X}} \mathrm{d}\mu(x)}, \tag{9}$$

$$q_p(d_a) = p\text{-quantile of } d_a(x), \tag{10}$$

$$C_{\max}^{(a)} = \text{largest connected high-defect cluster}, \tag{11}$$

$$\mathrm{Var}(d_a) = \left\langle (d_a - \langle d_a \rangle_\mu)^2 \right\rangle_\mu. \tag{12}$$

The multi-scale defect-field functional is then written as

$$\boxed{F_{\mathrm{multi}}[\Omega] = F_{\mathrm{mean}} + F_{\mathrm{tail}} + F_{\mathrm{cluster}} + F_{\mathrm{scale}}.} \tag{13}$$

The old scalar form is the special case in which only low-frequency mean supports are retained:

$$F_{\mathrm{macro}} = \mathcal{P}_{\mathrm{macro}}[F_{\mathrm{multi}}], \tag{14}$$

where $\mathcal{P}_{\mathrm{macro}}$ discards tails, clusters, and scale modes.

5 Multi-Scale Projection

5.1 Scale Operator

Let ℓ be a length, graph radius, redshift window, frequency shell, or coarse-graining scale. Define a scale projection

$$\boxed{d_a^{(\ell)}(x) = \Pi_\ell[d_a](x).} \tag{15}$$

Typical choices include:

- convolution or smoothing on lattices;
- graph-ball averaging over r-hop neighbourhoods;
- wavelet or spectral filtering;
- redshift/time binning;
- solver-trace windowing.

5.2 Defect Flow Across Scales

The scale dependence of defects can be described by an effective flow:

$$\boxed{\partial_{\log \ell} d_a^{(\ell)}(x) = \beta_a[d^{(\ell)}](x).} \tag{16}$$

This is not yet a renormalization group theorem. It is an operational definition of how structural defects change under coarse-graining.

The scalar macro-projection is valid only when this flow becomes self-averaging:

$$q_{0.95}\left(d_a^{(\ell)}\right) \approx \left\langle d_a^{(\ell)} \right\rangle \quad \text{and} \quad C_{\max}^{(a)}(\ell)/|\mathcal{X}| \ll 1. \tag{17}$$

SAT-like domains violate this condition through rare tails and connected frustration clusters.

6 Defect-Field Dynamics

Many current papers use defect fields diagnostically. The next step is to treat them as effective dynamical variables:

$$\boxed{\partial_t d_a(x,t) = \mathcal{F}_a[d](x,t) + \xi_a(x,t).} \tag{18}$$

Here $\mathcal{F}_a$ is a deterministic drift functional and ξ_a is an optional noise, driving, or environment term.

A minimal relaxation model is

$$\tau_a \partial_t d_a = -d_a + d_a^\star[\Phi, \mathcal{C}] + D_a \nabla_{\mathcal{X}}^2 d_a + \sum_b K_{ab} d_b, \tag{19}$$

where:

- $d_a^\star$ is the local target defect induced by the current domain;
- $D_a \nabla_{\mathcal{X}}^2 d_a$ is spatial/graph diffusion;
- K_{ab} captures cross-sector coupling.

This makes explicit what was previously only implicit: structural selection may be diagnostic, but structural defects can also have their own effective dynamics.

7 Robustness Closure

Robustness is not treated as a fifth primitive sector. At support level:

$$R_{\text{resp}}(x) = (H(x)B(x)V(x))^{1/3}, \tag{20}$$

and

$$R_{\text{rob}}(x) = \min\left[R_{\text{resp}}(x), (H(x)B(x)S(x)V(x))^{1/4}\right]. \tag{21}$$

For multi-scale systems, robustness can be projected at multiple resolutions:

$$R_{\text{rob}}^{(\ell)}(x) = \min\left[R_{\text{resp}}^{(\ell)}(x), \left(H^{(\ell)}B^{(\ell)}S^{(\ell)}V^{(\ell)}\right)^{1/4}\right]. \tag{22}$$

This avoids returning to an independent R field while allowing robustness to be local, scale-dependent, and domain-sensitive.

8 Observation and Projection Operators

Let Ω_{full} be the full structural domain and Ω_{obs} the observed or measured representation. An observation projection is

$$\boxed{\Pi_{\text{obs}} : \Omega_{\text{full}} \longrightarrow \Omega_{\text{obs}}.} \tag{23}$$

Examples include:

- field configuration $\to$ coarse observables;
- SAT formula graph $\to$ scalar feature vector;
- cosmological history $\to$ $H(z)$ or $f\sigma_8(z)$ samples;
- quantum dynamics $\to$ pointer-basis robustness scores;
- high-dimensional data $\to$ kNN graph statistics.

Projection loss can be measured by comparing two projections:

$$\boxed{\Delta_{\Pi}(\Omega) = \|\Pi_1(\Omega) - \Pi_2(\Omega)\|.} \tag{24}$$

This formalizes a recurring lesson: paradoxes, failures, and false universality claims often arise when a projection discards the structural mode that carries the relevant information.

9 Mode Separation

The v1.4.1 lesson is not that all sectors are mathematically orthogonal in every domain. It is that sector definitions must be operationally separated enough that different physical or computational situations do not collapse into the same structural coordinate. Let

$$\mathbf{g}(x) = (d_H(x), d_B(x), d_S(x), d_V(x)). \tag{25}$$

Define the sector covariance

$$C_{ab} = \text{Cov}_\mu(d_a, d_b). \tag{26}$$

Approximate mode separation means

$$\frac{|C_{ab}|}{\sqrt{C_{aa}C_{bb}}} \ll 1 \qquad (a \neq b), \tag{27}$$

or, more weakly, that a stable rotation exists:

$$\tilde{d}_i(x) = \sum_a U_{ia} d_a(x), \tag{28}$$

such that the modes $\tilde{d}_i$ have clearer transfer or prediction behaviour than the raw sectors.

This is the correct interpretation of Papers 52–55: some structural modes transfer, some remain domain-specific, and scalar universality is too strong.

9.1 v1.4.1 Anchor: Separating Balance and Activity

The concrete v1.4.1 correction was motivated by the B/S degeneracy exposed in the quantum pointer-state benchmark. There, balance and activity can be made operationally distinct by using

$$\boxed{B = 1 - |p_0 - p_1|, \qquad S = \frac{|\rho_{01}|}{\sqrt{\rho_{00}\rho_{11}}}.} \tag{29}$$

In this example, B measures population balance while S measures coherent off-diagonal activity. Thus the four combinations

$$(B, S) \in \{\text{low}, \text{high}\}^2 \tag{30}$$

represent physically distinct regimes:

Balance B	Activity S	Interpretation
low	low	imbalanced and incoherent
high	low	balanced but inactive/decohered
low	high	coherent but imbalanced
high	high	balanced coherent superposition/activity

The formulas above are domain-specific to the quantum density-matrix setting. Their general lesson is broader:

> *v1.4.1 requires sector definitions to separate structural modes operationally, not merely rename correlated summaries.*

10 Universality Criteria

The word "universal" should be reserved for no-retuning transfer. Let D_{train} and D_{test} be domains or benchmark families. Let θ denote a frozen structural architecture: sector definitions, summary operators, projection rules, and weights. Define a transfer gain

$$\Delta I(D_{\text{test}}; \theta) = I(S_\theta; y) - I(S_{\text{base}}; y), \tag{31}$$

where S_θ is the structural score or feature representation and S_{base} is a declared baseline.

A minimal universality criterion is

$$\boxed{U(\theta) = \mathbb{E}_{D_{\text{test}} \notin D_{\text{train}}} \left[\mathbf{1}\{\Delta I(D_{\text{test}}; \theta) > 0\} \right].} \tag{32}$$

This criterion is intentionally empirical. It requires:

- frozen definitions;
- no retuning on the target domain;

- declared baselines;
- shuffled or permuted nulls;
- reporting failures.

Paper 55 shows why this matters: mixed-family cross-validation can look positive while leave-family-out transfer fails. Therefore universality must be tested under domain shift.

11 Why This Formalisation Matters

The main gain is diagnostic precision. Earlier scalar scores could say that a system has high or low structural support, but they could not say where the relevant information was lost. The defect-field formulation separates four possibilities:

- the mean defect is already informative;
- rare high-defect tails dominate behaviour;
- connected defect clusters carry the signal;
- the relevant structure appears only after changing scale.

This makes failures interpretable. If two SAT families have similar mean supports but different CDCL hardness, the theory no longer has to treat this as a mysterious failure of "MAAT" as a whole. It can ask whether the missing information lives in propagation tails, local frustration clusters, or family-specific scale geometry. Likewise, if a smooth cosmological toy model is well described by a scalar support while SAT is not, the difference is not a contradiction. It means the two domains admit different valid projections of the same general defect-field language.

In this sense, v1.4.1 turns a broad question– "does one structural score transfer everywhere?"– into sharper questions: which modes transfer, at which scale, under which projection, and against which null model?

12 What Is Now Defined

Object	Definition introduced here
Local defect field	$d_a(x) = \mathcal{D}_a(\Phi\|_{\mathcal{N}(x)}, \mathcal{C}\|_{\mathcal{N}(x)}, \partial\mathcal{N}(x), \mu)$
Scale projection	$d_a^{(\ell)} = \Pi_\ell[d_a]$
Scale flow	$\partial_{\log \ell} d_a^{(\ell)} = \beta_a[d^{(\ell)}]$
Defect dynamics	$\partial_t d_a = \mathcal{F}_a[d] + \xi_a$
Observation projection	$\Pi_{\text{obs}} : \Omega_{\text{full}} \to \Omega_{\text{obs}}$
Projection loss	$\Delta_\Pi = \\|\Pi_1(\Omega) - \Pi_2(\Omega)\\|$
Universality score	$U = \mathbb{E}[\mathbf{1}\{\Delta I > 0\}]$ under no-retuning transfer

13 Limitations

- The operators are formal definitions, not microscopic derivations.
- The scale-flow equation is an effective coarse-graining description, not a proven renormalization group.
- Defect dynamics remain phenomenological until tied to domain-specific evolution laws.
- Universality is defined operationally through transfer tests, not asserted as a theorem.
- The framework still requires independent benchmarks beyond repository internal domains.

14 Conclusion

MAAT v1.4.1 should be read as a move from scalar scoring toward defect-field geometry. The important objects are no longer only global supports or one robustness number, but local defects, tails, clusters, scale projections, observation maps, and transfer criteria.

The conservative lesson is:

> *global structural scores are macro-projections; the full object is a projected, scale-dependent defect-field geometry.*

This does not prove universality. It makes universality testable by requiring frozen definitions, declared projections, shuffled nulls, and no-retuning transfer across new domains.

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